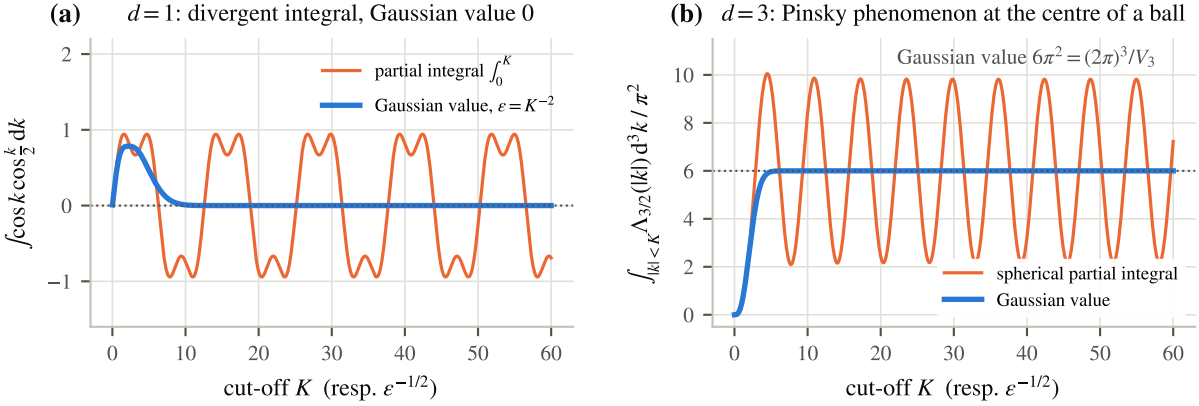
(a)
d = 1: divergent integral, Gaussian value 0
partial integral ∫_0^K
Gaussian value, ε = K^−2
∫ cos k cos^k/2 dk
cut-off K (resp. ε^−1/2)
(b)
d = 3: Pinsky phenomenon at the centre of a ball
Gaussian value 6π^2 = (2π)^3/V_3
spherical partial integral
Gaussian value
∫_|k|<K Λ^3/2 (|k|) d^3k / π^2
cut-off K (resp. ε^−1/2)